Emergent Gravity from Bandwidth Constraints in a Message-Passing Lattice

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We introduce message-passing lattices (MPLs) as a minimal computational substrate: nodes with bounded local state, connected by links of finite bandwidth, updated by a uniform local rule. Neighbor states are not instantly visible but must be transmitted via messages, and transmission time grows with message size.

This architecture poses a synchronization problem. Fully synchronous updates—where each node waits for all neighbors—allow a single slow region to freeze the entire network. Fully asynchronous updates—where nodes proceed independently—permit unbounded drift between regions. We propose bounded staleness as a middle path: each node adjusts its pace based on its own transmission delay and the average delay of its neighbors, controlled by a coupling parameter $\alpha \in [0, 1]$.

When local activity is high, state changes are large, messages are large, and links saturate. Saturated links cause delays. Delays spread through synchronization pressure. The result is a position-dependent reduction in update rate, which we describe by a scalar time-sag field $\phi(x) \leq 0$.

From the time-sag field, gravitational phenomenology follows: clocks in congested regions run slow (time dilation) and propagating patterns refract toward deeper ϕ (free fall). Numerical simulations on a 2D lattice confirm the predicted field profile, orbital motion, and flyby deflection.

The usual intuition treats gravity as a fundamental force that causes time dilation. Here the order is reversed: differential clock rates arise from bandwidth saturation, and gravitational attraction emerges as a consequence.

I. INTRODUCTION

Consider a network of nodes that communicate by passing messages. If neighbor states are not instantly visible but must be transmitted, and if links have finite capacity, a fundamental tension arises: how should nodes synchronize their updates?

A fully synchronous policy—where each node waits for all neighbors before proceeding—causes any local slowdown to propagate globally. A fully asynchronous policy—where nodes update independently—allows unbounded drift between regions. The middle ground is *bounded staleness*: nodes adjust their pace to limit drift from neighbors, without requiring lockstep synchronization.

In this paper we show that bounded staleness, combined with finite bandwidth, produces gravitational phenomenology. High-activity regions saturate their links, causing delays that spread through synchronization pressure. The result is a smooth, position-dependent reduction in local update rate—a scalar field we call the *time-sag*. In the weak-congestion regime, this field satisfies a Poisson equation sourced by activity density, with solutions matching the Newtonian gravitational potential.

The usual intuition treats gravity as a fundamental force that causes time dilation; here the order is reversed. Differential clock rates arise from bandwidth saturation, and gravitational attraction follows as a consequence.

The paper proceeds as follows. Section II defines the message-passing lattice substrate. Section III shows how space and time emerge from connectivity and update dynamics. Section IV introduces the synchronization problem and bounded staleness. Section V derives the time-sag field. Section VI develops gravitational phenomenology. Section VII presents numerical validation. Section VIII discusses extensions toward general relativity, and Section IX concludes.

II. MESSAGE-PASSING LATTICES

A. Architecture

A message-passing lattice (MPL) is a discrete computational substrate with one defining property: *neighbor states are not instantly visible*. Each node knows only its own state; to learn about neighbors, it must receive explicit messages. The basic structure is illustrated in Fig. 1.

Formally, an MPL consists of:

- A graph $G = (V, E)$ with vertices (nodes) $x \in V$ and edges (links) $e \in E$.
- A local state register s_x at each node x, with bounded size.
- Bidirectional links, each with finite capacity C bits per update opportunity.
- A uniform local update rule applied at every node.

Each node has bounded degree: $\deg(x) \leq d_{\max}$ for some fixed constant. This enforces locality. The graph need not be regular; irregularities can encode structure.

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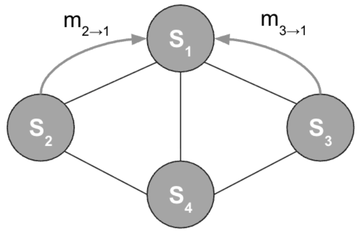

FIG. 1. A message-passing lattice. Each node x has local state s_x and updates based on incoming messages from neighbors. Neighbor states are not directly visible; information must be explicitly transmitted.

A cellular automaton is a special case of an MPL where nodes transmit their full state to all neighbors on every cycle. The MPL framework is more general: nodes may transmit partial information, compressed representations, or state deltas, and the cost of transmission becomes an explicit part of the dynamics.

B. Update Dynamics

The engine evolves through local update cycles. Each cycle at node x consists of two stages:

1. **Compute**: Read incoming messages and current state s_x. Apply the local transition rule to produce a new state s'_x and outgoing messages.

2. **Propagate**: Transmit a message m_x encoding the state change $\Delta s_x = s'_x - s_x$ to all neighbors.

The compute stage has fixed cost. The propagate stage does not: when the state change Δs_x is large, the message m_x is large, and transmission takes longer. This asymmetry between computation and communication is the source of all that follows.

III. EMERGENT GEOMETRY

A. Space from Connectivity

There is no background space. Spatial structure emerges from the connectivity of the graph.

We assume statistical regularity: the count of reachable nodes grows as h^D with hop count h, defining an effective dimension D. Effective distance between nodes is operational: the round-trip signaling time in completed cycles defines $d_{\mathrm{eff}}(x, y)$. At coarse scales, this behaves like an ordinary spatial metric. We do not assume any geometric embedding; the lattice *is* the space.

B. Time from Updates

There is no global clock. Nodes update asynchronously, and the engine's history is a partial order of local update events, not a sequence of global states.

The intrinsic time recorded by a node is its *proper time* τ_x: the number of update cycles that node x actually completes. A clock is any localized pattern that cycles through distinguishable states, advancing one step per completed update. Two clocks can compare their accumulated τ when they meet, but there is no substrate-level reference against which distant clocks are compared.

For analysis we introduce an external *coordinate time* t that counts *update opportunities*—steps of a fair scheduler that selects nodes to attempt updates. This t is not an observable of the MPL; it is a bookkeeping device used to define throughput constraints (C bits per opportunity) and completion rates:

$$f(x) \;=\; \frac{d\tau_x}{dt} \;\in\; [0, 1] \tag{1}$$

In a quiet region where every update opportunity completes, $f(x) = 1$. In a congested region where some cycles stall, $f(x) < 1$. The variation of $f(x)$ across the lattice is the origin of gravitational time dilation.

IV. THE SYNCHRONIZATION PROBLEM

A. Finite Bandwidth

Each link can carry at most C bits per update opportunity. What happens when this capacity is insufficient?

Nodes transmit state deltas Δs_x rather than full states. The size of the message $|m_x|$ scales with the magnitude of the state change. We define the *activity* $a(x)$ at node x as a measure of local dynamics—higher activity means larger state changes and larger messages. The transmission time for a message is:

$$\texttt{local\_delay}(x) \;=\; \frac{|m_x|}{C} \;\sim\; \frac{a(x)}{C} \tag{2}$$

When $|m_x| > C$, transmission cannot complete in one update opportunity and the node stalls.

B. Synchronous vs. Asynchronous Updates

How should nodes coordinate their updates? Two extremes bracket the design space.

Synchronous updates. Each node waits for all neighbors' messages before proceeding. This ensures consistency: every node sees fresh neighbor data. But it couples the entire lattice. If one node stalls—because its message is too large for the available bandwidth—its neighbors must wait. Their neighbors must wait for

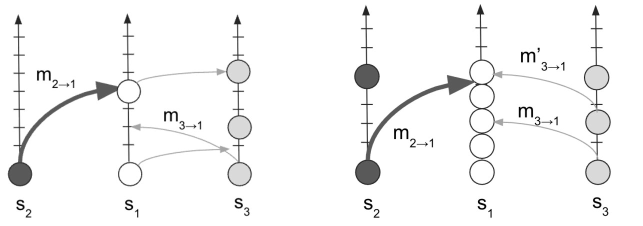

FIG. 2. Two synchronization extremes. (a) Synchronous: node s_1 waits for all neighbors before updating; a single slow node stalls the lattice. (b) Asynchronous: nodes update independently; fast nodes race ahead, leading to unbounded drift.

them. A single congested region propagates outward until the entire network runs at the pace of its slowest node.

Asynchronous updates. Each node proceeds independently, using whatever neighbor information is available. This avoids global stalls: fast regions race ahead while slow regions lag behind. But without any coupling, the drift between regions grows without bound. Nodes that should be neighbors in any meaningful sense may find themselves arbitrarily far apart in update count. Locality breaks down.

Neither extreme is satisfactory. Synchronous updates produce global fragility; asynchronous updates destroy coherent structure. These two extremes are illustrated in Fig. 2.

C. Bounded Staleness

The solution is *bounded staleness*: nodes adjust their pace to limit drift from neighbors, without requiring lockstep synchronization. This is not a new idea. Analogous mechanisms appear throughout distributed systems:

- *Backpressure* in network protocols, where producers slow down when consumers cannot keep up.
- *Bounded staleness consistency* in distributed databases, which caps drift between replicas without enforcing synchronous commits.
- *Gradient clock synchronization*, which minimizes clock skew between nearby nodes in a network.

The common thread is local feedback: each node adjusts its pace based on information from immediate neighbors, and global coordination emerges without central control.

Many rules could implement bounded staleness. We adopt a simple one. Let $\langle \texttt{delay} \rangle_x$ be the average delay among neighbors of node x, inferred from gaps between received messages. The node schedules its next update as:

$$\texttt{next\_update}(x) \;=\; \max\!\Big(\texttt{local\_delay}(x),\; \alpha \cdot \langle \texttt{delay} \rangle_x\Big) \tag{3}$$

where $\alpha \in [0, 1]$ controls coupling strength.

The limiting cases recover our extremes:

- $\alpha = 0$: No coupling. Nodes proceed based on local delay alone. Fully asynchronous.
- $\alpha = 1$: Full coupling to neighbor average. A node slows to match the mean pace of its neighborhood.

Intermediate values $0 < \alpha < 1$ damp the coupling: nodes feel their neighbors but the influence is attenuated. As we show in the next section, the choice of α has profound consequences for the range of the emergent interaction.

V. THE TIME-SAG FIELD

A. Steady-State Completion Rate

Consider a lattice with a static activity distribution $a(x)$. In steady state, each node settles into a completion rate $f(x) \in [0, 1]$: the fraction of update opportunities that result in completed cycles.

From Eq. (3), the delay at node x is the maximum of its local delay and α times the average neighbor delay. In steady state, delay and completion rate are inversely related: $\texttt{delay}(x) \propto 1/f(x)$. Writing $\lambda(x) = 1 - f(x)$ for the stall fraction (small in the weak-congestion regime), the synchronization rule becomes a self-consistency condition:

$$\lambda(x) \;=\; \max\!\Big(\lambda_{\text{local}}(x),\; \alpha \cdot \langle \lambda \rangle_x\Big) \tag{4}$$

where $\lambda_{\text{local}}(x) \sim \gamma\, a(x)$ is the stall fraction from local bandwidth saturation and $\langle \lambda \rangle_x$ is the average stall fraction among neighbors of x.

B. Coupling Regimes

In the weak-congestion limit ($\lambda \ll 1$), the max softens to a linear combination, and the self-consistency condition can be analyzed via the graph Laplacian (see Appendix A). The result depends critically on α:

Damped coupling ($\alpha < 1$). The steady-state equation takes the form of a *screened Poisson equation*:

$$\nabla^2 \lambda \;-\; k^2 \lambda \;=\; -\kappa\, \rho_{\text{act}} \tag{5}$$

where the screening wavevector $k \propto \sqrt{1-\alpha}$ sets an exponential decay scale. Solutions fall off as e^{-kr}/r in three dimensions—a Yukawa potential. Synchronization pressure dissipates at each hop; the interaction is short-ranged.

Full coupling ($\alpha = 1$). The screening term vanishes, leaving a pure Laplacian:

$$\nabla^2 \lambda \;=\; -\kappa\, \rho_{\text{act}} \tag{6}$$

This is Poisson's equation. Solutions fall off as $1/r$ in three dimensions (or $\log r$ in two dimensions)—long-range, like Newtonian gravity.

The transition at $\alpha = 1$ is not gradual: any $\alpha < 1$ produces exponential screening, regardless of how close to unity. Long-range behavior, characteristic of Newtonian gravity, emerges only when synchronization pressure propagates without attenuation. In this sense, full coupling is not merely a convenient choice but a necessary condition for the gravitational analogy to hold.

C. Definition of the Time-Sag Field

We define the *time-sag field* as the deviation of the completion rate from its uncongested value:

$$\phi(x) \;\equiv\; f(x) - 1 \;=\; -\lambda(x) \tag{7}$$

With this sign convention:

- $\phi(x) = 0$ in quiet regions (full completion, no congestion).
- $\phi(x) < 0$ in congested regions (reduced completion rate).

Active regions sit in potential wells, with ϕ becoming more negative where congestion is worse. The field satisfies $\phi(x) \in [-1, 0]$, with $\phi = -1$ representing complete congestion.

For full coupling ($\alpha = 1$), the time-sag field satisfies:

$$\nabla^2\phi \;=\; \kappa\,\rho_{\mathrm{act}} \tag{8}$$

This is mathematically identical to the Newtonian gravitational potential, with activity density ρ_{act} playing the role of mass density. For a localized source in three dimensions, $\phi \propto -1/r$. In two dimensions, $\phi \propto -\log r$.

VI. GRAVITATIONAL PHENOMENOLOGY

A. Time Dilation

A clock is a localized pattern whose internal state cycles through distinguishable phases, advancing one step per completed update. The accumulated step count is the proper time τ_x at node x.

In a quiet region where $f(x) = 1$, the clock advances once per update opportunity. In a congested region where $f(x) < 1$, some update opportunities fail to complete and the clock advances more slowly. Since $\phi(x) = f(x) - 1$, the proper time evolves as:

$$\frac{d\tau}{dt} = 1 + \phi(x) \tag{9}$$

Clocks in congested regions ($\phi < 0$) run slow relative to clocks in quiet regions ($\phi = 0$). This is gravitational time dilation: clocks near massive bodies—here, regions of high activity—tick at a reduced rate. The effect arises not from any explicit time-warping mechanism but from the local failure of update cycles to complete.

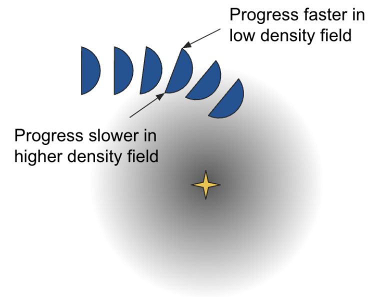

FIG. 3. Refraction of a propagating pattern in the time-sag field. The portion of the wavefront nearer the source (deeper ϕ) completes fewer updates and falls behind, tilting the wavefront and bending the trajectory inward.

B. Free Fall

Patterns propagate through the lattice by successive state updates along connected nodes. In a region of uniform ϕ, wavefronts advance symmetrically and trajectories are straight. When ϕ varies, the portion of a wavefront in deeper ϕ (more congested) completes fewer updates per unit coordinate time, falls behind, and the trajectory bends toward the slower region (Fig. 3).

This is analogous to Fermat's principle: light refracts toward regions of lower propagation speed. Here, patterns refract toward regions of lower completion rate—toward deeper ϕ. The deflection depends only on the local gradient of ϕ and the pattern's velocity, not on its internal structure or amplitude. A small perturbation and a large wavepacket follow the same trajectory through a given field.

VII. NUMERICAL VALIDATION

To test these predictions, we developed an open-source simulator implementing bandwidth-limited message-passing dynamics.[1]

A. Simulation Setup

The simulator models a 2D square lattice with absorbing boundary conditions (boundary nodes update at the uncongested rate without delays). Each node x carries local state, tracks completed update cycles, and implements the synchronization rule of Eq. (3) with $\alpha = 1$. Links have uniform capacity C, and activity $a(x)$ is specified as a static spatial distribution.

[1] Available at `https://github.com/eschnou/mpl-universe-simulator`

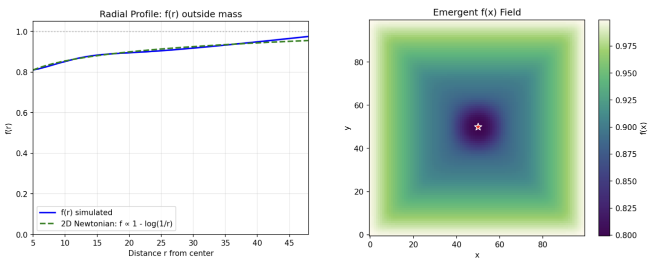

FIG. 4. Steady-state time-sag field for a single localized source on a 2D lattice. (a) Radial profile of $f(r)$ outside the source, compared to the 2D Newtonian prediction $f \propto 1 - \log(1/r)$. (b) Emergent $f(x)$ field around the source (star). Darker regions indicate lower completion rate.

A localized mass is modeled as a circular region of elevated activity surrounded by a quiet background ($a = 0$). The simulation runs until the completion rate field $f(x)$ converges to steady state, from which we compute $\phi(x) = f(x) - 1$.

B. Radial Profile

For a single localized source, the steady-state time-sag field is approximately radially symmetric. Figure 4 shows the azimuthally averaged $f(r)$ as a function of distance from the source center.

In two dimensions, the Poisson equation yields a logarithmic potential: $\phi \propto -\log(r/r_0)$ outside the source. The simulated profile matches this prediction, confirming that the synchronization dynamics produce the expected long-range field. Deviations near the source boundary reflect the finite size of the active region and discretization effects.

C. Orbital Motion

To test free fall, we inject a propagating particle into the converged ϕ field and track its trajectory. The particle is initialized with a velocity perpendicular to the radial direction from the source. The resulting trajectories orbit the active region, consistent with motion in an attractive central potential (Fig. 5). In a 2D logarithmic potential, orbits generically precess rather than close; our simulations exhibit this expected behavior.

D. Flyby Deflection

A more direct test of the effective force law is gravitational deflection. We initialize a particle on a trajectory that would, in the absence of the source, pass at some closest-approach distance $r_{\min}$. We then measure the asymptotic deflection angle between incoming and outgoing directions. Figure 6 shows a flyby with deflection angles consistent with scattering in a logarithmic potential.

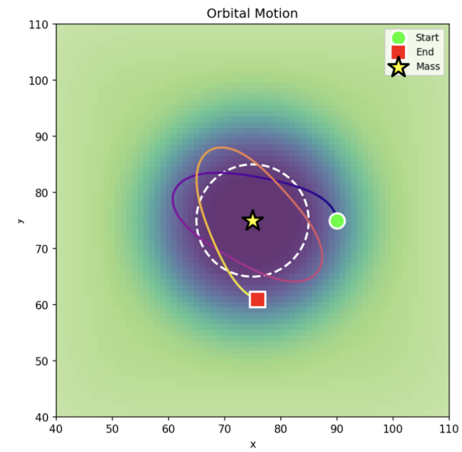

FIG. 5. Orbital motion of a propagating particle around a localized activity source. In a 2D logarithmic potential, orbits precess rather than close.

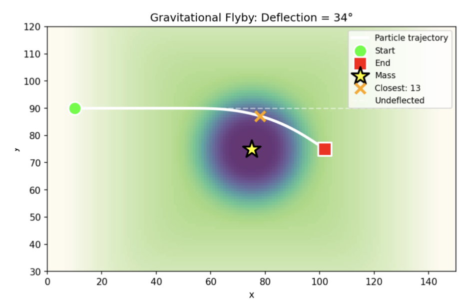

FIG. 6. Gravitational flyby deflection. A particle passing the source at closest approach $r_{\min} \approx 13$ lattice units is deflected by approximately 34. Dashed line shows the undeflected trajectory.

E. Limitations

The current implementation is two-dimensional, so potentials are logarithmic rather than $1/r$. Extension to three dimensions is straightforward in principle but computationally demanding.

The activity distribution is static and externally imposed. The simulations therefore validate that imposed activity produces the predicted field and dynamics, but do not yet demonstrate self-consistent sourcing by propagating patterns. Whether a wavepacket can generate sufficient activity to measurably influence other patterns remains to be tested.

VIII. TOWARD GENERAL RELATIVITY

The time-sag field $\phi(x)$ is a scalar. This suffices for Newtonian gravity but not for general relativity, which requires a rank-2 metric tensor.

Our derivation tracks only total activity at each node—a single number. But if the update rule consults a multi-hop neighborhood, the engine naturally sees not just how much activity there is, but how it is distributed spatially and whether information flows asymmetrically.

This suggests a route to richer structure:

- *Total activity* sources the time-sag field (gravitational time dilation).
- *Directional flow*—asymmetric bandwidth usage in different directions—might source frame-dragging effects.
- *Anisotropic congestion* could provide the spatial metric components.

We have not derived the Einstein equations, but some of the architectural ingredients are present. Developing this into a full tensorial theory is left for future work.

IX. CONCLUSION

We have shown that gravitational phenomenology can emerge from a minimal computational substrate operating under finite bandwidth.

The core mechanism is a synchronization problem. In a message-passing lattice, nodes must coordinate their updates despite having no global clock. Fully synchronous updates cause global fragility; fully asynchronous updates allow unbounded drift. Bounded staleness—where nodes adjust their pace to limit drift from neighbors—provides a middle path. When combined with finite link capacity, this synchronization pressure transforms local congestion into a smooth, lattice-wide slowdown field.

The resulting time-sag field $\phi(x)$ satisfies a Poisson equation sourced by activity density, yielding $1/r$ potentials in three dimensions and logarithmic potentials in two. Clocks in congested regions tick slowly, and propagating patterns refract toward regions of deeper ϕ. These are the signatures of gravitational time dilation and free fall, arising from bandwidth constraints rather than spacetime curvature.

A key finding is that long-range behavior requires full coupling: any damping of the synchronization pressure ($\alpha < 1$) converts the Poisson equation into a screened Poisson equation, producing exponentially decaying Yukawa potentials. Newtonian gravity emerges only at $\alpha = 1$, where synchronization pressure propagates without attenuation.

Numerical simulations on a 2D lattice validate these predictions. The steady-state ϕ field around a localized activity source shows the expected logarithmic profile. Propagating particles exhibit orbital motion and flyby deflection consistent with an attractive central potential.

Several questions remain open. The current simulations impose activity externally; whether propagating patterns can self-consistently source the fields they traverse—so that two wavepackets attract each other through the congestion they each generate—has not been demonstrated. Extension to three dimensions would test whether the mechanism scales correctly. And the connection between activity in the MPL and physical mass or energy remains to be established.

Nonetheless, the central result stands: a discrete network with bounded communication, operating under a simple local synchronization rule, produces a long-range attractive interaction with the characteristic features of Newtonian gravity. The usual intuition treats gravity as a fundamental force that causes time dilation; here the order is reversed. Time dilation arises first, from bandwidth saturation, and gravitational attraction follows as a consequence. This suggests that gravity-like behavior may be a natural consequence of resource constraints on any local, finite substrate.

Appendix A: Derivation of the Time-Sag Field Equation

1. Setup

We derive the steady-state behavior of the time-sag field from the synchronization rule introduced in Section IV. Consider an MPL where each node x has activity $a(x)$ and each link has capacity C bits per update opportunity.

2. The Micro-Rule

Each node schedules its next update according to:

$$\texttt{next\_update}(x) = \max\big(\texttt{local\_delay}(x),\ \alpha \cdot \langle \texttt{delay} \rangle_x\big) \tag{A1}$$

where:

- $\texttt{local\_delay}(x) = |m_x|/C \sim \gamma\, a(x)$ is the time to transmit the outgoing message, proportional to activity with constant $\gamma > 0$.
- $\langle \texttt{delay} \rangle_x$ is the average delay among neighbors of x, inferred from gaps between received messages.
- $\alpha \in [0, 1]$ is the coupling strength.

3. Steady-State Condition

In steady state, each node settles into a constant delay $\delta(x) = \texttt{next\_update}(x)$. The completion rate $f(x)$—the fraction of update opportunities that result in completed cycles—is inversely related to delay:

$$f(x) = \frac{1}{\delta(x)} \tag{A2}$$

where we normalize so that $f = 1$ corresponds to one completion per opportunity (no stalling).

Define the stall fraction $\lambda(x) = 1 - f(x)$, which is small in the weak-congestion regime. Since $f(x) = 1/\delta(x)$, we have $\delta(x) = 1/(1 - \lambda(x))$, which for small λ expands to:

$$\delta(x) \approx 1 + \lambda(x) \tag{A3}$$

4. Linearized Self-Consistency

Substituting into Eq. (A1), the steady-state delay satisfies:

$$\delta(x) = \max\bigl(\gamma\, a(x) + 1,\ \alpha \cdot \langle \delta \rangle_x\bigr) \tag{A4}$$

where we have written $\texttt{local\_delay}(x) = \gamma\, a(x) + 1$ to account for the baseline delay of one opportunity even with zero activity.

In the weak-congestion regime, where the local and neighbor contributions are both small perturbations, the max operation can be approximated by a sum (both contributions independently push the delay above baseline). Subtracting 1 from both sides and using $\lambda = \delta - 1$:

$$\lambda(x) \ \approx\ \gamma\, a(x) \ +\ \alpha \cdot \langle \lambda \rangle_x \tag{A5}$$

This is the self-consistency condition: a node's stall fraction arises from local bandwidth saturation (first term) and coupling to neighbor stall fractions (second term).

5. Laplacian Form

Rearranging Eq. (A5):

$$\lambda(x) - \alpha \cdot \langle \lambda \rangle_x = \gamma\, a(x) \tag{A6}$$

On a regular lattice with degree d, the graph Laplacian acts as:

$$(L\lambda)(x) = \sum_{y \sim x} [\lambda(x) - \lambda(y)] = d\,[\lambda(x) - \langle \lambda \rangle_x] \tag{A7}$$

Solving for the neighbor average:

$$\langle \lambda \rangle_x = \lambda(x) - \frac{1}{d}(L\lambda)(x) \tag{A8}$$

Substituting into the self-consistency equation:

$$\lambda(x) - \alpha \left[\lambda(x) - \frac{1}{d}(L\lambda)(x)\right] = \gamma\, a(x) \tag{A9}$$

This simplifies to:

$$(1 - \alpha)\,\lambda(x) + \frac{\alpha}{d}(L\lambda)(x) = \gamma\, a(x) \tag{A10}$$

6. Coupling Regimes

The character of Eq. (A10) depends on α:

a. No coupling ($\alpha = 0$). The Laplacian term vanishes:

$$\lambda(x) = \gamma\, a(x) \tag{A11}$$

Stalling is purely local. Each node's congestion depends only on its own activity, with no spatial propagation.

b. Damped coupling ($0 < \alpha < 1$). Both terms contribute. Passing to the continuum limit (see below), Eq. (A10) becomes a screened Poisson equation:

$$\nabla^2 \lambda - k^2 \lambda = -\kappa\, \rho_{\rm act} \tag{A12}$$

where the screening wavevector is:

$$k = \frac{1}{\ell}\sqrt{\frac{d(1-\alpha)}{\alpha}} \tag{A13}$$

and ℓ is the lattice spacing. Solutions decay exponentially beyond the screening length $\xi = 1/k$. For α close to but below unity, $\xi \propto 1/\sqrt{1-\alpha}$ diverges, but remains finite.

c. Full coupling ($\alpha = 1$). The first term in Eq. (A10) vanishes, leaving:

$$(L\lambda)(x) = d\gamma\, a(x) \tag{A14}$$

This is a pure Laplacian equation with no screening. Solutions are long-ranged.

7. Continuum Limit

On a regular D-dimensional lattice with spacing ℓ, the graph Laplacian approaches the continuum Laplacian:

$$(L\lambda)(x) \to -\ell^2 \nabla^2 \lambda \tag{A15}$$

The sign arises because the graph Laplacian is positive when a node exceeds its neighbors, while ∇^2 is negative at a local maximum.

For full coupling ($\alpha = 1$), Eq. (A14) becomes:

$$\nabla^2 \lambda = -\kappa\, \rho_{\rm act} \tag{A16}$$

where $\rho_{\rm act}(x)$ is the coarse-grained activity density and $\kappa > 0$ absorbs γ, d, and ℓ.

8. Conversion to Time-Sag

Defining the time-sag field $\phi = -\lambda = f - 1$, we obtain:

$$\nabla^2 \phi = \kappa\, \rho_{\rm act} \tag{A17}$$

This is Poisson's equation with source $\rho_{\rm act}$. For a localized source in three dimensions, $\phi \propto -1/r$; in two dimensions, $\phi \propto -\log(r/r_0)$. These match the Newtonian gravitational potential, with activity density playing the role of mass density.